

The empirical recurrence for OEIS A222434 is false, and the correct one is the same line with one more term and a range

Adrian Perez Fontelles
Independent researcher

22 September 2026

Abstract

OEIS A222434 carries an empirical recurrence of order 25 contributed by R. H. Hardin. It is false: the entry's b-file runs to $n = 210$, so the recurrence can be tested at 185 indices and it fails at every one, the first being $n = 26$ where it falls short by 3. A transfer matrix on 62 states, reproducing all 210 b-file terms, settles both halves. Appending a single term $-a(n - 26)$ gives a recurrence that the annihilation test certifies for every $n > 54$, while the published line is certified for no n at all. The range is not decoration: the corrected line also fails at eight indices below 55, so this is not the published line with a term accidentally omitted — it is a different statement.

2020 Mathematics Subject Classification. 05A15, 11B37, 15B36.

1 The sequence and the claim

OEIS A222434 is “Number of binary arrays indicating the locations of trailing edge maxima of a random length- n 0..4 array extended with zeros and convolved with 1,1,1.”. It has offset 1, publishes 38 terms and links a b-file of 210. It carries one formula line, an empirical recurrence of order 25:

$$\begin{aligned} a(n) = & a(n-1) + a(n-2) - a(n-3) + a(n-4) + a(n-5) - a(n-6) \\ & + a(n-7) + a(n-8) - a(n-9) + a(n-10) + a(n-11) - a(n-12) \\ & + 2a(n-13) - a(n-14) - a(n-15) + a(n-16) - a(n-17) - a(n-18) \\ & + a(n-19) - a(n-20) - a(n-21) + a(n-22) - a(n-23) - a(n-24) \\ & + a(n-25), \quad (1) \end{aligned}$$

recorded as empirical on the live entry (revision 8, Oct 2025), with no range stated and nothing recording it as settled or corrected.

With offset 1 and order 25 the line first asserts something at $n = 26$. DATA stops at $n = 38$, so the DATA alone tests it thirteen times; the b-file tests it 185.

2 The published recurrence fails at all 185 indices its b-file can test

n	$a(n)$	(1) predicts	difference
26	194714	194711	+3
27	314846	314848	-2
28	509094	509097	-3
29	823186	823187	-1
30	1331062	1331067	-5
31	2152279	2152288	-9

Proposition 1. (1) is false. It fails at every one of the 185 indices $n = 26, \dots, 210$ at which the entry's b-file can test it; the smallest is $n = 26$, where $a(26) = 194714$ and (1) gives 194711.

Proof. Direct evaluation in exact integer arithmetic on the 210 integers the entry's b-file publishes. It is worth noting that thirteen of these failures are already visible in the entry's own DATA section, without the b-file. $\square$

3 The model

The counting problem is a walk count: the condition defining a trailing edge maximum of the convolved array is local, and a state carrying enough of the recent window determines which extensions are admissible. Built and reduced, the automaton has $S = 62$ states, and $a(n) = \mathbf{v}^\top M^n \mathbf{u}$ reproduces *all* 210 b-file terms exactly — every one, not a sample — and hence all 38 published DATA terms.

4 The recurrence that does hold

Theorem 2. For every $n > 54$,

$$a(n) = [\text{the right-hand side of (1)}] - a(n - 26). \quad (2)$$

Proof. Let q be the characteristic polynomial of (2) and $w = q(M)\mathbf{u}$. For n beyond the order, $a(n) - \sum_i c_i a(n-i) = \mathbf{v}^\top M^{n-55} w$, so the recurrence holds from there exactly when $\mathbf{v}^\top M^m w = 0$ for every $m \geq 0$; by Cayley–Hamilton, checking $m < S = 62$ suffices. Carried out in exact integer arithmetic, the residuals vanish and the test returns the threshold 54. $\square$

Proposition 3. (1) is certified for no threshold at all: the same test applied to the published coefficients returns none, so there is no index beyond which (1) holds.

Remark 4. The stated range in Theorem 2 is not a formality. Evaluated on the b-file, (2) fails at exactly eight indices below 55 — $n = 29, 35, 38, 41, 45, 48, 51, 54$ — and holds at all 156 indices from $n = 55$ to $n = 210$. So (2) is not (1) with a term accidentally dropped in transcription; it is a different statement, true only eventually, and a reader who appended $-a(n - 26)$ to the published line and tested it from $n = 27$ would conclude it was false too.

5 Verification

Four checks.

First, Proposition 1 uses only the 210 integers of the entry's own b-file, and thirteen of its failures need only the DATA. It does not depend on the model.

Second, the model reproduces all 210 b-file terms, and therefore all 38 DATA terms.

Third, (2) was checked directly against the b-file at every index from $n = 27$ to $n = 210$: it fails at the eight indices listed above and holds at all 156 others, matching Theorem 2's threshold of 54 exactly.

Fourth, both lines were put through the same annihilation test in the same code: it certifies (2) from $n = 55$ and returns nothing for (1).

Remark 5. This is the fourth entry of this shape settled here in one day, after A269637 (order 10 published, 13 true), A236647 (34, 38) and A196074 (37, 43). In the first three the published coefficients are the true recurrence's first coefficients term for term, and the correction is a pure truncation. This one is the variant worth separating: the published coefficients are again a prefix, but what is missing is one term *and* a range, and without the range the corrected line is false as well.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A222434.
- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [3] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011.
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.